Chapter 13

Vectors I

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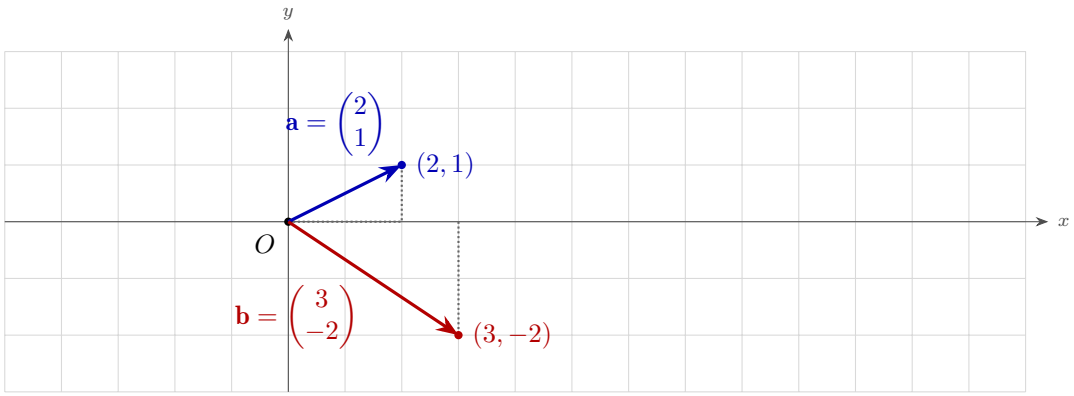
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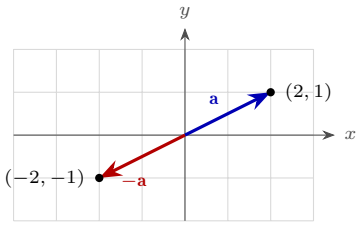
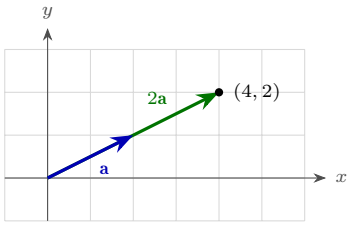
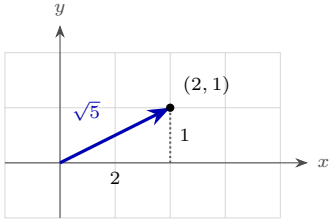
13.1 Vectors in Two Dimensions

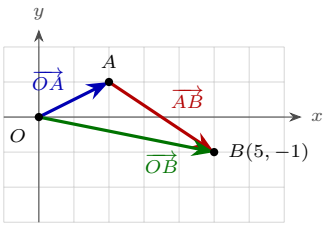
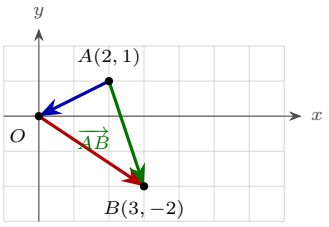
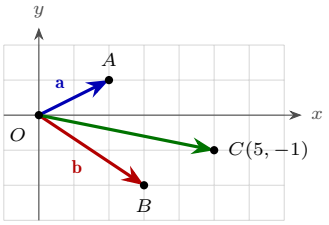
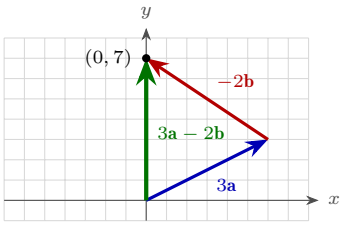
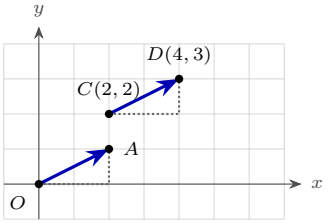
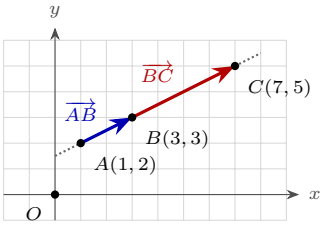
In Secondary 4, we have learned vectors in two dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$$

A vector describes a quantity that has both magnitude and direction. Vectors let us describe position, displacement and direction algebraically, so that geometric problems can be solved by calculation. In H2 Mathematics, we extend vectors from two dimensions to three dimensions, and use them to work with lines and planes, angles, distances, intersections and the foot of the perpendicular. Outside mathematics, vectors are used for forces and velocities in physics, in navigation, in computer graphics and 3D modelling, and throughout engineering. Vectors also lie at the heart of mathematics in linear algebra, which has foundational applications in fields such as physics, engineering, computer science, data science, machine learning and artificial intelligence, economics and cryptography.



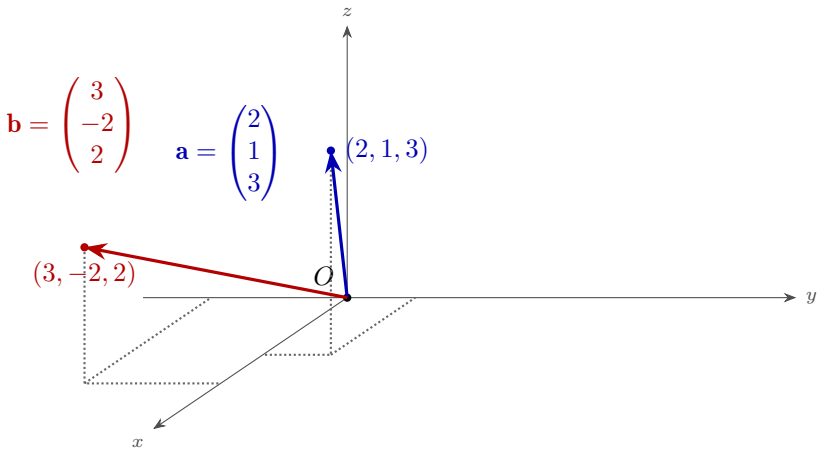
Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$	
2**a**	$k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \end{pmatrix}\right\rvert = \sqrt{x^2 + y^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2 + 1^2}$ $= \sqrt{5}$	

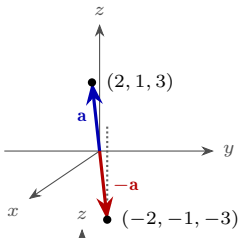
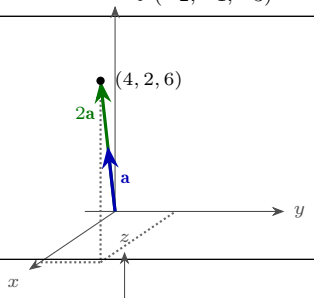
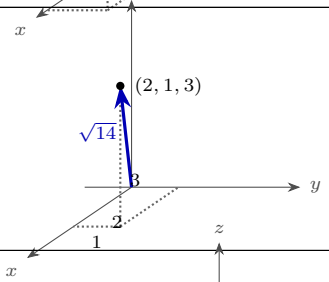
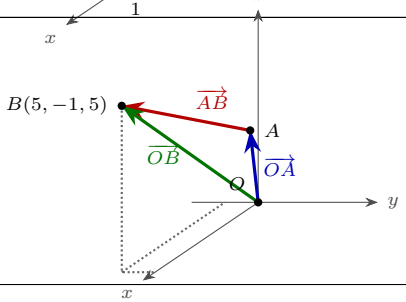
Name	Equation	Diagram
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 4-2 \\ 3-2 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 3-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 7-3 \\ 5-3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

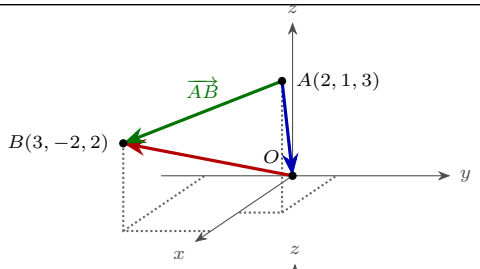
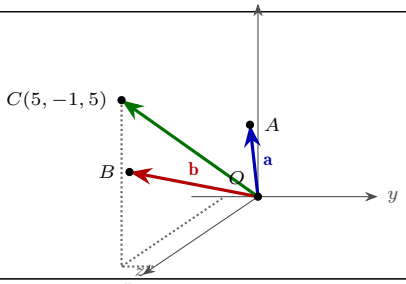
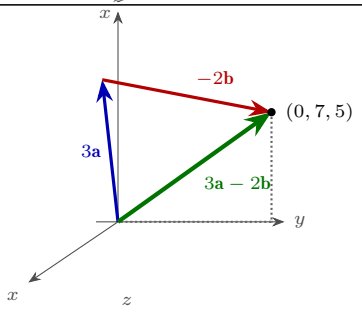
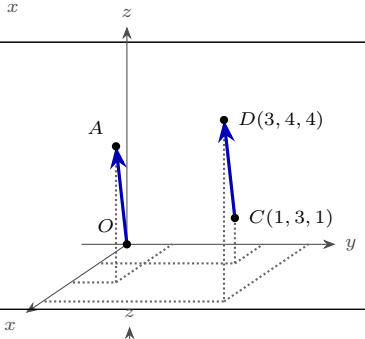
13.2 Vectors in Three Dimensions

In H2 Mathematics, we extend vectors to three dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$



Name	Equation	Diagram
Negative of $\mathbf{a}$ $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \\ -3 \end{pmatrix}$	
2a	$k\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} kx \\ ky \\ kz \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\rvert = \sqrt{x^2+y^2+z^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2+1^2+3^2}$ $= \sqrt{14}$	
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	

Name	Equation	Diagram
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \\ \lambda a_3 + \mu b_3 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 3-1 \\ 4-3 \\ 4-1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 2-1 \\ 3-2 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 4-2 \\ 5-3 \\ 7-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$ $= 2\overrightarrow{AB}$	